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RESEARCH ARTICLE

# Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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## Abstract

The ICH E17 guideline describes regional pooling in multi-regional clinical trials (MRCTs) in terms of the similarity of effect modifier distributions, but provides no quantitative methodology. We propose a planning-stage framework that quantifies effect modifier dissimilarity using the Wasserstein-1 ($W_1$) distance, retained in the original measurement units of each effect modifier. Unlike the standardized mean difference (SMD), the $W_1$ distance captures differences in scale and skewness in addition to location—distributional features that may drive treatment effect heterogeneity through non-linear effect modification. Because it requires only baseline effect modifier distributions, $W_1$ can be computed from any representative data source, including prior trials, disease registries, and real-world evidence (RWE) databases—enabling a common planning scenario in which a small-sample region identifies suitable pooling partners before the trial begins. We develop a dual-pathway clinical calibration: when prior evidence on conditional average treatment effect (CATE) sensitivity ($L$) is available, the $W_1$ distance is translated into a maximum potential regional treatment effect difference ($\Delta_{\max}$) with bootstrap confidence intervals; when $L$ is unknown, the framework reverse-calculates the required CATE sensitivity ($L^*$) at pre-specified clinical margins. Simulation studies across seven systematic scenarios demonstrated satisfactory bias and coverage of the percentile bootstrap estimator for non-negligible distributional differences at moderate sample sizes per region. We illustrate the framework through a hypothetical thrombolytic MRCT grounded in publicly available GUSTO-I individual patient data, with one region designated as a small-sample anchor and the remaining regions evaluated as pooling partners on two candidate effect modifiers (age and systolic blood pressure). Partner rankings differed markedly between the two effect modifiers, with several partners ranked similar on one but dissimilar on the other, demonstrating that all candidate effect modifiers must be evaluated jointly. A small subset of partners emerged as leading pooling candidates because they ranked low on both candidate effect modifiers and required clinically implausible CATE sensitivity to produce a meaningful treatment effect difference. The principal contribution of this work is to transform the previously qualitative judgment of "similar enough" into a reproducible quantitative basis for sponsor judgment, filling the methodological gap in ICH E17 implementation and supporting evidence-based regional pooling decisions from any representative population data source.

**KEYWORDS**
multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

## 1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become the standard paradigm for global pharmaceutical development.[1,2] The International Council for Harmonisation (ICH)

**Abbreviations:** AMI, acute myocardial infarction; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; EDF, empirical distribution function; EU, European Union; GUSTO, Global Utilization of Streptokinase and t-PA for Occluded coronary arteries; ICH, International Council for Harmonisation; IPD, individual patient data; IQR, interquartile range; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; NMPA, National Medical Products Administration; RWE, real-world evidence; SBP, systolic blood pressure; SMD, standardized mean difference; US, United States.

  

E17 guideline, adopted in 2017, established principles for planning and designing MRCTs to evaluate whether treatment effects can be regarded as generalizable across the target population.[3]

Regulatory authorities in Japan and China expect demonstration of treatment effect consistency in regional subpopulations,[4,5] yet individual regional sample sizes in global MRCTs are often insufficient for such demonstration. A key strategy for addressing this challenge is the regional pooling approach, wherein regions with similar patient characteristics are grouped for analysis to enhance the ability to assess regional consistency.[3] The ICH E17 guideline describes pooling decisions in terms of the similarity of effect modifier distributions.[3]

An effect modifier is a baseline patient characteristic—such as age, disease severity, or genetic marker—for which the treatment benefit differs across subgroups. When such heterogeneity exists, even if the drug works identically at the individual level, regions with different patient compositions may observe different average treatment effects. A region with predominantly younger patients would show larger benefits than a region with predominantly older patients, not because the drug works differently, but because the patient mix differs. This fundamental relationship underscores why the similarity of the full effect modifier distribution, not only its average level, is critical to the validity of regional pooling. In this paper, we focus on continuous effect modifiers, such as age and disease severity scores; categorical effect modifiers raise distinct methodological considerations and are outside the present scope.

The extent to which the relevant effect modifiers are known varies across development programmes, and this shapes what can be planned quantitatively.[6] For some treatments, mechanism or prior trials identify a specific effect modifier in advance—a genetic marker known to determine response, for instance—so the characteristic requiring cross-regional comparison is established before planning. More often at the planning stage, the effect modifiers are only partially understood, and the sponsor proceeds from candidate characteristics suggested by biological plausibility or earlier-phase data, or from broader regional and ethnic factors that stand in as surrogates for the underlying modifiers.[6] The framework developed here operates on whichever set of candidate effect modifiers the sponsor specifies: it quantifies and clinically calibrates the cross-regional distributional similarity of each candidate, and is agnostic to how the candidates were identified.

Despite the regulatory importance of effect modifier distributional similarity, current practice lacks a standardized, clinically calibrated methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are "similar enough." Song et al., writing from the China NMPA perspective on ICH E17 implementation, note the challenge of operationalizing pooling criteria without quantitative tools.[5] The most concrete existing proposal appears in the dedicated pooling-strategy chapter of a reference volume on multi-regional trials under ICH E17, where Komiyama et al. operationalize pooling through cluster analysis on region-level effect modifier summaries: each region is assigned a single representative value for each candidate effect modifier—their worked example uses the proportion of male and of younger patients—and inter-regional Euclidean distances in that space are weighted for relevance via the Lasso.[6] This proposal leaves two gaps that we address. First, because each regional distribution enters only through one summary per modifier, two regions that agree on the chosen summary but differ in spread, shape, or tail behaviour are at distance zero; more generally, any finite summary requires specifying in advance which distributional features are relevant, rather than comparing the full distributions directly. Second, it explicitly calls for a judgment of whether effect modifier differences are "clinically meaningful or not" but supplies no metric to operationalize that judgment.

This absence of quantitative tools is particularly acute at the planning stage, when sponsors must identify suitable pooling partners for small-sample regions. A workshop report by Matsushima et al. documented that regional imbalance in CRP-positive/MRI-negative status contributed to apparent inconsistency in the secukinumab MRCT.[4] This example illustrates the practical consequences when effect modifier distributional differences are not assessed quantitatively at the planning stage. That case involved a binary status, where imbalance reduces to a difference in proportions; the continuous effect modifiers on which we focus can differ in shape and spread, not only average level, so assessing their similarity requires more than comparing means.

General-purpose comparison tools such as the standardized mean difference are routinely applied to baseline covariates,[7] but the standardized mean difference captures only differences in location: regions can match on means yet differ in variance or distributional shape. When effect modification is non-linear, these features directly influence regional average treatment effects.

This paper addresses these gaps by proposing a quantitative framework for assessing effect modifier distributional similarity across regions, based on the Wasserstein-1 ($W_1$) distance between two cumulative distribution functions, retained in the original measurement units of the effect modifier. Geometrically, $W_1$ equals the area between the two cumulative distribution functions. Unlike the standardized mean difference, $W_1$ responds to differences in variance and shape as well as location. We estimate $W_1$ with bootstrap confidence intervals and link it to clinical interpretation through a calibration framework.

The remainder of this paper is organized as follows. Section 2 presents the methodological framework, including the index definition, clinical calibration, and estimation procedure. Section 3 describes a simulation study. Section 4 demonstrates the framework using publicly available individual patient data. Section 5 discusses implications, limitations, and future directions.

## 2 | METHODS

An effect modifier is a factor for which the treatment effect varies across subgroups. Formally, let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where $X$ is the effect modifier. When this function is non-constant, the average treatment effect observed in region $r$ depends on the distribution $F_r$ of the effect modifier in that region:

$$\bar{\tau}_r = \int \tau(x)\, dF_r(x). \tag{1}$$

Assessing pooling suitability therefore requires a measure of distributional difference between regions.

### 2.1 | Existing Approaches and Their Limitations

The most widely used measure for comparing covariate distributions across groups is the standardized mean difference (SMD), defined as

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_2}{S_{\text{pooled}}}, \tag{2}$$

where $\bar{X}_r$ denotes the sample mean in region $r$ and $S_{\text{pooled}}$ is the pooled standard deviation.[7] Because treatment effects in clinical trials are typically summarized as mean differences, SMD provides a scale-free comparison directly relevant to the quantities that drive clinical decision-making. However, by construction SMD captures only differences in location. Two distributions with identical means but markedly different variances—for example, $N(50, 5^2)$ and $N(50, 15^2)$—yield SMD = 0 despite a threefold difference in spread.

Distributional distance measures address this limitation by comparing distributions beyond summary statistics. Empirical distribution function (EDF) tests such as the Kolmogorov–Smirnov (KS) statistic, $D_{\text{KS}} = \sup_x |F_1(x) - F_2(x)|$, and related measures (Cramér–von Mises, Anderson–Darling) quantify differences in the full shape of distributions by comparing CDFs directly. Information-theoretic divergences such as Kullback–Leibler (KL) divergence,

$$D_{\text{KL}}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx, \tag{3}$$

take a density-based approach, measuring how one probability distribution differs from another.

However, each of these approaches has limitations when applied to comparing effect modifier distributions across regions. SMD captures only location differences and is blind to variance and shape. EDF-based statistics such as the KS statistic capture full distributional differences but lack a clinically interpretable scale and have no theoretical connection to treatment effect heterogeneity. KL divergence is asymmetric—comparing region A to region B yields a different value than comparing B to A—can diverge to infinity when empirical distributions do not fully overlap, and requires density estimation that is impractical with small regional samples. Crucially, none of these measures offers a theoretical link that would translate a distributional distance into a bound on regional treatment effect differences. These gaps motivate a measure that (i) captures distributional features beyond location, (ii) provides a clinically interpretable scale, and (iii) offers such a theoretical link. The Wasserstein-1 distance and its clinical calibration, developed in the following subsections, satisfy all three requirements.

### 2.2 | Per-EM Wasserstein-1 Distance

Among distributional distances that could capture the full distributional differences relevant to regional pooling, we adopt the Wasserstein-1 distance ($W_1$) for its unique theoretical connection to treatment effect heterogeneity (Proposition 1 below). The $W_1$ distance (sometimes called the Earth Mover's Distance) admits three equivalent representations on the real line, each highlighting a different facet of its interpretation.[8,9]

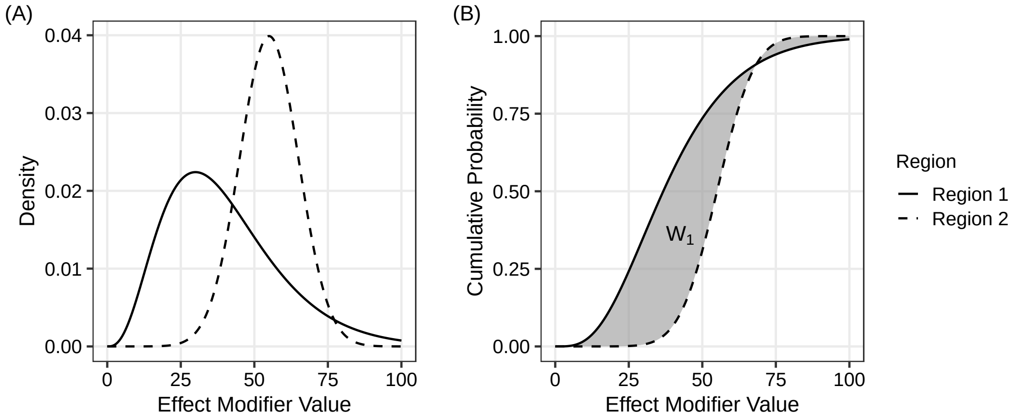


**FIGURE 1** The Wasserstein-1 distance as the area between cumulative distribution functions, illustrated with a right-skewed Region 1 (gamma distribution) and a symmetric Region 2 (normal distribution). Panel (A): probability densities. Panel (B): cumulative distribution functions; the shaded area equals $W_1(F_1, F_2)$.

Let $F_1$ and $F_2$ denote the cumulative distribution functions (CDFs) of an effect modifier in two regions. Then

$$W_1(F_1, F_2) = \int_{-\infty}^{\infty} |F_1(x) - F_2(x)|\, dx = \int_0^1 |F_1^{-1}(u) - F_2^{-1}(u)|\, du = \inf_{\pi \in \Pi(F_1, F_2)} \int |x - y|\, d\pi(x, y), \tag{4}$$

where $F_r^{-1}$ is the quantile function and $\Pi(F_1, F_2)$ is the set of probability couplings with marginals $F_1$ and $F_2$. The CDF form expresses $W_1$ as the area between the two CDFs (Figure 1); the quantile form as the integrated absolute difference of quantile functions; and the Kantorovich form as the minimum expected absolute displacement to transport one distribution onto the other. The three forms are mathematically equivalent.[9] Unlike the standardized mean difference, $W_1$ responds to differences in variance, skewness, and other distributional features.[10]

A property of $W_1$ central to our framework is that it carries the units of the underlying effect modifier. In each of the three equivalent forms in equation (4), the integrand has units of the variable $X$: the CDF values are dimensionless probabilities, the quantile function $F^{-1}$ returns values in the original measurement units, and the coupling integral sums distances $|x - y|$ in those units. Consequently, $W_1(F_1, F_2)$ is expressed on the original scale of the effect modifier. This dimensional alignment supports direct clinical interpretation: a regional $W_1 = 5$ mmHg in systolic blood pressure means that an expected 5 mmHg of probability mass must be transported between the two distributions to align them. The same logic precludes a meaningful direct comparison of $W_1$ values across effect modifiers measured on different scales, and we therefore restrict attention in this paper to per-effect-modifier assessment (Section 5 elaborates on this scope choice).

$W_1$ is a proper metric on the space of probability distributions with finite first moment, satisfying non-negativity, symmetry, identity of indiscernibles, and the triangle inequality.[8] It is also affine-equivariant in the variable units, satisfying $W_1(F_{aX+b}, F_{aY+b}) = |a| \cdot W_1(F_X, F_Y)$ for any $a \neq 0$. Beyond these basic properties, $W_1$ admits a Kantorovich–Rubinstein dual representation[8]

$$W_1(F_1, F_2) = \sup_{\|f\|_{\mathrm{Lip}} \leq 1} \left| \int f\, dF_1 - \int f\, dF_2 \right|, \tag{5}$$

which expresses $W_1$ as the largest possible difference in expected values of a 1-Lipschitz function $f$ between the two distributions (the notation $\|f\|_{\mathrm{Lip}} \leq 1$ means $|f(x) - f(y)| \leq |x - y|$ for all $x, y$). If the conditional average treatment effect function $\tau(x)$ is Lipschitz continuous with constant $L_{\mathrm{clinical}}$, so that $|\tau(x) - \tau(y)| \leq L_{\mathrm{clinical}}|x - y|$ for all $x, y$, then $\tau/L_{\mathrm{clinical}}$ is 1-Lipschitz and applying equation (5) yields the central result of this section:

**Proposition 1.** (Connection to heterogeneity). *If the CATE function is Lipschitz continuous with constant $L_{clinical}$, then for any two regions with effect modifier distributions $F_1$ and $F_2$,*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L_{clinical} \cdot W_1(F_1, F_2). \tag{6}$$

*Proof.* Apply equation (5) with $f = \tau/L_{\text{clinical}}$, which is 1-Lipschitz by assumption. This yields $\left|\int(\tau/L_{\text{clinical}})\,d(F_1 - F_2)\right| \leq W_1(F_1, F_2)$. Combining with equation (1) for $\bar{\tau}_r$ and rearranging yields equation (6). □

The bound in equation (6) is specific to $W_1$. The $W_2$ distance, while admitting closed-form expressions for Gaussian families, does not possess a Lipschitz dual representation and therefore cannot provide an analogous bound. The alternative measures discussed in Section 2.1 similarly lack any analogous property.[11] Its right-hand side involves $W_1$ in its original measurement units, multiplied by a clinical slope $L_{\text{clinical}}$; no further normalization of $W_1$ is required to obtain a clinically interpretable bound. We therefore adopt $W_1$ in its original units as the primary similarity metric throughout this paper. Some prior literature has considered normalizing $W_1$ by a within-effect-modifier scale measure such as the pooled interquartile range; we show in Section 2.4 and Supplement A that such normalization is mathematically redundant once equation (6) is used for clinical calibration. The Lipschitz assumption underlying Proposition 1 is appropriate when the CATE function varies smoothly with the effect modifier; for threshold or step-function CATEs, $L_{\text{clinical}}$ can be arbitrarily large and the bound becomes conservative (Section 5).

## 2.3 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, the Wasserstein-1 distance is estimated using empirical distribution functions. Since $\hat{F}_1$ and $\hat{F}_2$ are piecewise constant (right-continuous step functions), the integral in equation (4) reduces to a finite sum over the intervals defined by the combined order statistics:

$$\widehat{W}_1 = \sum_{k=1}^{n_1+n_2-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)}), \tag{7}$$

where $\hat{F}_r$ denotes the empirical CDF (right-continuous) of region $r$ and $x_{(1)} < \cdots < x_{(n_1+n_2)}$ are the combined order statistics. The estimator $\widehat{W}_1$ inherits the original measurement units of the effect modifier. The asymptotic distribution of $W_1$ is non-standard: del Barrio et al.[12] showed that $\sqrt{n}\,W_1(\hat{F}_n, F) \xrightarrow{d} \int |B(F(x))|\,dx$, where $B$ is a standard Brownian bridge. Because the quantiles of this limiting distribution depend on the unknown $F$, no universal critical values are available and direct construction of asymptotic confidence intervals is impractical. We therefore employ the nonparametric percentile bootstrap with $B = 2{,}000$ replicates.[13] In the two-sample setting, when $F_1 \neq F_2$ the Hadamard derivative of the $L_1$ functional is linear, so the ordinary nonparametric bootstrap is consistent.[14] Convergence occurs at rate $\sqrt{n_1 n_2/(n_1 + n_2)}$ (see Appendix A.2 for details). The estimator (7) is computed in $O((n_1 + n_2)\log(n_1 + n_2))$ time, dominated by sorting the combined order statistics.

## 2.4 | Interpretation and Clinical Calibration

A central feature of $W_1$ is its connection to treatment effect heterogeneity through Proposition 1. This connection provides the basis for a clinically grounded interpretation framework rather than reliance on fixed numerical cutoffs.

We refer to the Lipschitz constant of the CATE function $\tau(x)$ as the clinical slope $L_{\text{clinical}}$. By definition, $L_{\text{clinical}}$ quantifies the maximum change in the treatment effect per unit change in the effect modifier, expressed on the outcome scale of the trial—for example, percentage points of 30-day mortality per year of age, or percentage points per mmHg of systolic blood pressure. When $L_{\text{clinical}}$ is estimable from prior data, individual patient data (IPD) meta-regression on treatment-by-covariate interaction provides a per-unit slope directly,[15,16] and the same quantity is denoted the interaction slope in the methodological literature.[17,18] More commonly at the MRCT planning stage, however, published subgroup reports document the direction and significance of interaction rather than a per-unit slope, and a precise numerical $L_{\text{clinical}}$ is unavailable. We address both situations below.

From Proposition 1, the maximum potential difference in regional average treatment effects attributable to effect modifier distributional differences is

$$\Delta_{\max} = L_{\text{clinical}} \cdot W_1(F_1, F_2), \tag{8}$$

expressed in the units of the trial outcome scale. The formula is invariant to within-effect-modifier scale normalization of $W_1$: as shown in Supplement A, any normalized form $W_1/X$ multiplied by the same scale $X$ cancels in equation (8), so adopting $W_1$ in its original units yields the same clinical-scale calibration as any rescaled version. When $L_{\text{clinical}}$ is independently known, equation (8) provides a direct clinical-scale translation of the observed distributional difference. When $L_{\text{clinical}}$ is unknown, the relation can be inverted to ask which $L_{\text{clinical}}$ would be required for a given $W_1$ to produce a clinically meaningful $\Delta_{\max}$.

The estimator $\widehat{W}_1$ has a percentile bootstrap CI (Section 2.3), and inferential uncertainty propagates linearly to $\Delta_{\max}$ when $L_{\text{clinical}}$ is treated as fixed: $[\Delta_{\max,L}, \Delta_{\max,U}] = L_{\text{clinical}} \cdot [W_{1,L}, W_{1,U}]$. When $L_{\text{clinical}}$ itself is uncertain or unavailable, we solve equation (8) for $L_{\text{clinical}}$ at a pre-specified clinical margin $\Delta_{\text{clin}}$ (the minimal clinically important treatment effect difference, the non-inferiority margin, or the targeted overall effect), yielding the required clinical slope

$$L^* = \frac{\Delta_{\text{clin}}}{W_1(F_1, F_2)}. \tag{9}$$

$L^*$ is the smallest clinical slope under which the observed distributional difference would produce a clinically meaningful regional treatment effect difference. If a plausible upper bound on $L_{\text{clinical}}$ from prior evidence (denoted $L_{\text{UB}}$) lies below $L^*$, the distributional difference is unlikely to translate into clinical heterogeneity, regardless of the unknown true $L_{\text{clinical}}$. The two pathways—forward calibration via equation (8) when $L_{\text{clinical}}$ is known and reverse comparison via equation (9) against $L_{\text{UB}}$ when $L_{\text{clinical}}$ is unknown—accommodate the full range of evidence states encountered at the MRCT planning stage.

Specifying $L_{\text{clinical}}$ (or its upper bound $L_{\text{UB}}$) draws on disease-area evidence. For the GUSTO-I case study presented in Section 4, the Fibrinolytic Therapy Trialists' meta-analysis[19] and GUSTO-I subgroup analyses[20,21] together support a conservative upper bound of $L_{\text{UB,age}} \approx 1 \times 10^{-2}$ per year and $L_{\text{UB,SBP}} \approx 2 \times 10^{-3}$ per mmHg on the 30-day mortality scale. "Conservative" here indicates that the chosen bounds exceed the empirical prognostic age slope of approximately $6 \times 10^{-3}$ per year reported by Lee et al.,[21] so that any plausible CATE sensitivity falls below them. Supplement D discusses the derivation of these bounds, alternative specification strategies such as the deft estimator of Fisher et al.,[16] and the limitations of class-level evidence. The methodology does not require an exact $L_{\text{clinical}}$; sensitivity analyses across a plausible range of values, or comparison of $L^*$ to $L_{\text{UB}}$, accommodate the typical uncertainty in this parameter.

The pooling assessment proceeds by computing per-effect-modifier $W_1$ with bootstrap CIs across candidate partner regions, deriving $\Delta_{\max}$ when $L_{\text{clinical}}$ is known or $L^*$ at a pre-specified $\Delta_{\text{clin}}$ when $L_{\text{clinical}}$ is unknown, and ranking partners by these quantities. The framework is deliberately estimation-centered rather than testing-based: $W_1$ has no universal cutoff that separates "similar" from "dissimilar," and the appropriate threshold depends on the clinical slope $L_{\text{clinical}}$, which is external to the distributional comparison. By presenting per-pair $\Delta_{\max}$ (or $L^*$) with bootstrap CIs alongside the clinical context—the overall treatment effect size, the non-inferiority margin, the minimal clinically important difference—the framework provides quantitative inputs for sponsor and regulatory judgment. The decision is not driven by a significance test[22]: where a screening rule is required, the clinical margin supplies it. Specifying a margin $\Delta_{\text{clin}}$ and an upper bound $L_{\text{UB}}$ induces the threshold

$$\tau_{\text{clin}} = \frac{\Delta_{\text{clin}}}{L_{\text{UB}}}, \tag{10}$$

the largest $W_1$ at which the distributional difference remains clinically tolerable, expressed in the units of the effect modifier. The threshold is therefore a property of the clinical margin rather than of the distance.

Deriving $\tau_{\text{clin}}$ leaves the calibration half done. A threshold is a usable decision rule only if the estimator can resolve differences on its scale, and $\widehat{W}_1$ cannot resolve arbitrarily small ones. That question is taken up next.

## 2.5 | Operability: The Null Floor

Because $W_1$ is a distance, $\widehat{W}_1$ is non-negative by construction and returns a positive value even when the two regions are draws from the same distribution. The relevant quantity is therefore not a bias to be subtracted but the sampling distribution of $\widehat{W}_1$ under the null $F_1 = F_2$, whose upper tail marks the smallest difference the estimator can distinguish from none at the sample sizes actually available. We refer to this distribution as the null floor, and to the comparison of a threshold against it as an operability check.

A threshold rule can discriminate only if its threshold lies above that floor. Writing $q_{1-\alpha}(n_1, n_2, F)$ for the $(1-\alpha)$ quantile of the null distribution of $\widehat{W}_1$ at the two regional sample sizes and the shape $F$ of the effect modifier, the necessary condition is

$$\tau_{\text{clin}} > q_{1-\alpha}(n_1, n_2, F). \tag{11}$$

When it fails, an "eligible" verdict is not evidence of similarity: it is compatible with the two regions being identical, and equally compatible with a difference the data cannot see. The level $\alpha$ is left free here; the choice made for the case study is stated in Section 4.

The check is inexpensive. Draw two independent resamples, of the two regions' actual sizes, from a single region's empirical distribution; recompute $\widehat{W}_1$; repeat. Because the floor depends on both sample sizes, it is computed per candidate partner rather than once for the trial. No additional data and no distributional assumption are required beyond those already used to estimate $W_1$, and the computation takes seconds.

This condition is not a restatement of a minimum sample size per region. A sample-size recommendation calibrated to bias and coverage (Section 3.2.2) asks whether $\widehat{W}_1$ estimates $W_1$ well; equation (11) asks whether the clinical margin is coarse enough for the estimator's resolution, and therefore depends on the strictness of $\Delta_{\text{clin}}$ and the shape of the effect modifier, not on $n$ alone. The two can diverge: in the case study of Section 4.2 the regional samples exceed two thousand observations, far above any sample-size recommendation we would make, and the operability condition nonetheless fails for one of the two candidate effect modifiers. Which modifier carries a pooling decision is thus a question the data answer, and answer differently from the question of whether the estimator is well behaved.

# 3 | SIMULATION STUDY

We conducted two complementary simulation studies. Study 2, presented first, evaluates decision performance: whether each candidate distance measure, used to select pooling partners, identifies the regions that truly share the anchor's effect modifier distribution. Study 1 evaluates estimation performance: bias, variability, and coverage probability of bootstrap confidence intervals for $\widehat{W}_1$. Partner selection is the deliverable of the framework, which is why Study 2 carries the primary comparison; Study 1 underwrites the calibration pathway that converts an estimated distance into a $\Delta_{\max}$ statement via equation (8).

Because the two studies answer different questions, they produce different sample-size statements, and a third arises from the operability condition of Section 2.5. Study 1's recommendation of $n \geq 100$ per region is an estimation criterion, calibrated to bias, coverage, and CI width. Study 2's required sample sizes are decision-accuracy criteria: for selection, the sample size at which a measure's discrimination reaches AUC $\geq 0.9$, which for $\widehat{W}_1$ ranges from 42 to 690 per region across the decisive cells reported below; for the clustering illustration, ARI $\geq 0.8$, with task floors of 36–376. The operability condition compares the estimator's null floor with the clinically derived threshold, and depends on the strictness of the clinical margin as well as on $n$. The three statements are labelled throughout and are not interchangeable.

## 3.1 | Study 2: Decision Performance

### 3.1.1 | Design

Study 2 follows the ADEMP structure for simulation studies.[23] Four worlds are simulated, each built from a single distributional family and each containing regions constructed either from the anchor's own population (true matches) or from a discordant population. Set 1 (Gaussian) places discordance in location and scale. Set 2 (log-normal) places it in location and in dispersion combined with shape. Set 3 (mixture) is the adversarial world for summary-based measures: every region, matched or discordant, is constructed with mean 50 and SD 10, so all discordance lies in distributional shape. Set 4 (extremes) displaces rare subgroups: 5% of a discordant region's population is moved 40–100 units away while the mean is held fixed, the pattern of a small severe subpopulation; one control region instead shifts its whole population by a small amount, a configuration deliberately favourable to CDF-supremum statistics. Ground truth is the construction label—which regions were built from the anchor's population—so the target of the decision is defined without reference to any of the compared measures and the comparison is not circular.

Six measures enter the comparison: $W_1$, the KS statistic, and SMD, together with three representative-value distances. RV1 is the representative-value distance of the pooling strategy of Komiyama et al. as written, with one representative value per effect modifier—for a continuous modifier, the regional mean—standardized across the roster and compared by Euclidean distance.[6] RV2 and RV3 are our extensions, not part of the cited proposal: they augment the coordinate vector with the regional SD (RV2) and additionally skewness (RV3), and are included to test whether added distributional summaries close the gap to full-distribution measures. On the log-normal world, SMD computed on the log scale is also run.

Two tasks score the measures against the same ground truth. In the selection task, an anchor and nine candidate regions are drawn; each measure ranks the candidates by their distance to the anchor and admits the $k$ closest, where $k$ is the number of true matches. Giving every measure the same $k$ removes threshold choice from the comparison entirely; the clinically derived threshold that $W_1$ alone possesses is exercised in the application (Section 4). Performance is summarized by the AUC for discriminating true matches from discordant candidates (chance = 0.5), by the false-pooling rate among the $k$ admitted, and by harm, defined as the expected largest true $W_1$ inside the selected pool—by Proposition 1, the worst-case regional treatment-effect difference the selected pool admits, per unit of $L_{\text{clinical}}$. In the clustering task, twelve regions in three true groups of four are partitioned from each measure's distance matrix (average linkage) and scored by the adjusted Rand index (ARI; chance = 0), first with the true number of clusters supplied and then with it estimated. The selection task ran 10,000 replications per cell at the production grid ($n$ = 25–100 per region) and 3,000 at the extended grid ($n$ = 150–2,000); the clustering task, 5,000 and 2,000. The extension to $n$ = 2,000 is what makes the classification below possible.

### 3.1.2 | Three Behaviours: Blind, Partial, and Recovering

Extending the sample-size grid to $n$ = 2,000 separates the measures into three behaviours (Figure 2). A measure is blind in a cell when its AUC is flat at chance through $n$ = 2,000: the population value of its distance carries no information about the construction label, so the task is unsolvable for that measure at any sample size. This is an identification failure, not a power failure. In Set 3 it is exact by construction: all regions there share mean and SD, so the population distance between representative-value coordinates built from the mean, or from mean and SD, is zero. On the selection task, RV1, RV2, and SMD are blind in the Set 3 combined cell and SMD in the Set 4 symmetric-severity cell (AUC 0.49–0.51 across the full grid); in the Set 3 shape-symmetric cell, where skewness is also matched, RV3 joins them.

A measure is partial when its AUC plateaus above chance but below the accuracy target: part of the structure is resolved, and the remainder never is. RV1 in the Set 4 symmetric-severity cell plateaus at 0.63, and RV3 in the Set 3 combined cell at 0.78. Neither is blind—both resolve real structure—but no sample size completes either.

A measure is recovering when its AUC rises monotonically through the target: identification succeeds, and what remains is a required-sample-size question. $W_1$ and the KS statistic are recovering in every world on both tasks. In particular, the KS statistic is blind nowhere: in the Set 4 cell where it is weakest it still reaches ARI 0.985 at $n$ = 2,000. Its failures in what follows are failures of efficiency, not of identification.

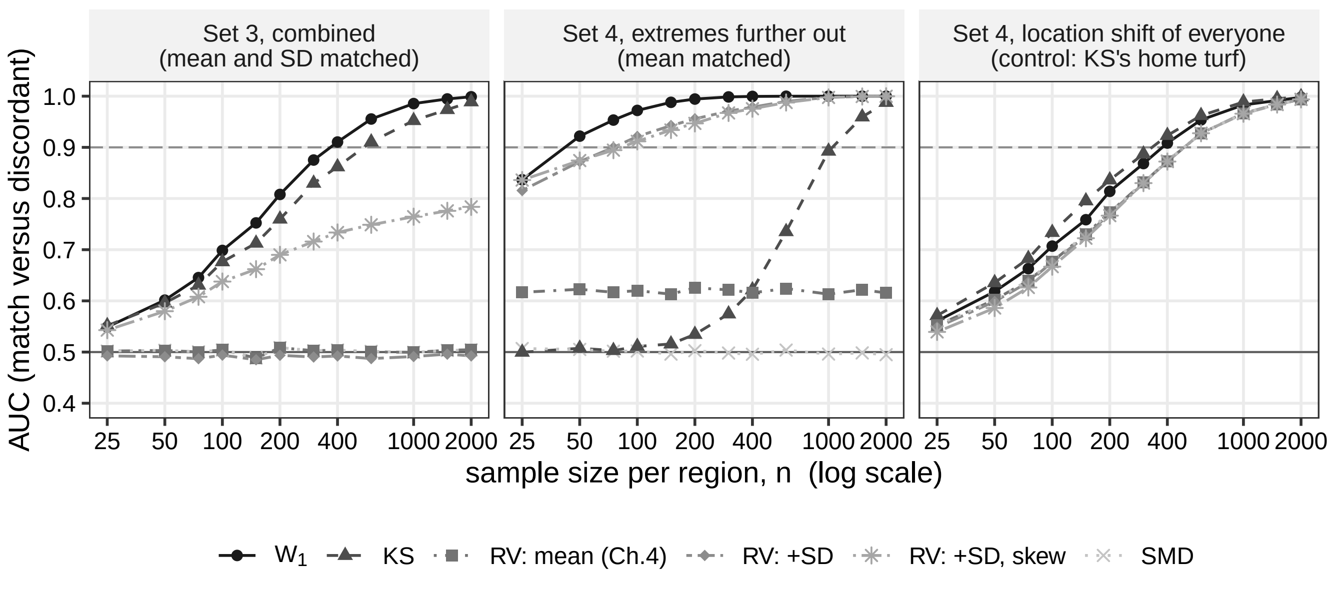


**FIGURE 2** Selection AUC (true matches versus discordant candidates) as a function of per-region sample size ($n$ = 25–2,000, log scale) for six distance measures in three cells of Study 2: the Set 3 combined cell (mean- and SD-matched mixtures), the Set 4 symmetric-severity cell (mean-matched displaced extremes), and the Set 4 bulk-shift control cell (uniform location shift of the whole population). The dashed horizontal line marks AUC = 0.9; chance corresponds to 0.5.

### 3.1.3 | Selection: Required Sample Sizes and the Cost of Errors

Table 1 reports, for the decisive cells, the per-region sample size at which each measure's selection AUC reaches 0.9. Two kinds of number appear in it, and they answer different questions. Reading down the $W_1$ column gives the difficulty of the task, from 42 (symmetric severity) to 690 (shape-symmetric mixtures). Reading across a row gives the method difference within one task: in the symmetric-severity cell, $W_1$ reaches the target at $n$ = 42 and the KS statistic at $n$ = 1,047, a 25-fold difference on the same data-generating process. The control cell behaves as designed: a uniform shift of the whole population is the configuration the CDF supremum is best suited to, the KS statistic wins it (333 versus 377), and we report the loss. Nor is the comparison uniformly favourable to $W_1$ against the coordinate extensions: in the asymmetric-severity cell, RV3 reaches the target at $n$ = 41, before $W_1$ at 67.

**TABLE 1** Per-region sample size at which selection AUC reaches 0.9, by cell and measure. Blind cells (AUC 0.49–0.51 across the full grid to $n$ = 2,000) are marked "blind"; partial cells (AUC plateaus above chance but below 0.9) are marked "partial" with the plateau in parentheses.

| Cell | $W_1$ | KS | SMD | RV1 | RV2 | RV3 |
|---|---|---|---|---|---|---|
| Set 4, symmetric severity | 42 | 1,047 | blind | partial (0.63) | 76 | 83 |
| Set 4, asymmetric severity | 67 | 1,252 | 530 | 389 | 125 | 41 |
| Set 4, bulk shift (control) | 377 | 333 | 489 | 491 | 490 | 491 |
| Set 3, combined | 368 | 550 | blind | blind | blind | partial (0.78) |
| Set 3, shape-symmetric | 690 | 716 | blind | blind | blind | blind |

Cells are contrast types within each world; "combined" pools a world's contrast types. In the shape-symmetric cell, RV2 and RV3 sit at or slightly below chance through $n$ = 2,000. Numeric entries in the RV2 and RV3 columns of the blind rows would be misleading and are replaced by the behaviour label.

The required-$n$ comparison counts errors; weighting them by their clinical cost widens the gap. At $n$ = 100 in Set 4 (all contrast types pooled), $W_1$'s false-pooling rate among the $k$ admitted is 0.718 against the KS statistic's 0.971, a 1.35-fold difference; harm—the expected largest true $W_1$ inside the selected pool—is 2.000 against 4.27, a 2.14-fold difference. The KS statistic errs more often, and it errs in the more expensive direction, admitting the regions whose true distances are largest. Across the four worlds at $n$ = 100, $W_1$ has the smallest harm in three (in the Gaussian world RV2 is smallest, 0.336 against $W_1$'s 0.402).

One further failure is a property of population values rather than of sample size. In Set 4, the true anchor distances of four discordant regions are, on the $W_1$ scale, 3.0, 6.0, 3.0, and 2.0; on the KS scale, 0.047, 0.050, 0.050, and 0.072. The orderings reverse: the clinically mildest region ($W_1$ = 2.0, the bulk-shift region) is the most discordant on the KS scale, and the clinically worst ($W_1$ = 6.0) is separated from the mildest displaced-extremes region by a KS gap of 0.003. Consequently, for any threshold $\tau$ between 3 and 6, no KS threshold admits exactly the regions with true $W_1 \leq \tau$—at any sample size, however the threshold is chosen. Where the clinical requirement is a bound on $W_1$-scale discrepancy, as the calibration of Section 2.4 makes it, the KS statistic cannot express the requirement even asymptotically.

### 3.1.4 | Clustering as a Second Scoring Currency

The classification of Section 3.1.2 was stated on the selection task. If it reflected an interaction between a measure and one task, a different task could rank the measures differently; if the defect lies upstream in the distance itself, an independent scoring currency should reproduce it. The clustering task provides that currency, and reproduces the classification cell for cell (Table 2). The blind measures remain at chance at $n$ = 2,000 (ARI 0.001–0.023 for RV1 and SMD in Sets 3 and 4, and 0.014 for RV2 in Set 3). The partial measures plateau (RV1 0.495 and SMD 0.460 in Set 1; RV2 0.380 and RV3 0.247 in Set 4). $W_1$ and the KS statistic recover everywhere, and the KS statistic again pays a large efficiency price where the CDF discrepancy is spread thin (required $n$ for ARI $\geq$ 0.8: 1,240 against $W_1$'s 376 in Set 4, and 629 against 364 in Set 3) while winning outright in Set 2 at moderate sample sizes (ARI 0.850 against 0.642 at $n$ = 100; required $n$ 83 against 156). The task floors—the best any measure achieves—are 36, 83, 364, and 376 across the four worlds, and the floor-achieving measure changes with the world (RV2 in Set 1, the KS statistic in Set 2, $W_1$ in Sets 3 and 4), which guards the comparison against the reading that it was constructed for one winner. The table also shows that adding moments is not free: in Set 1, where the mean and SD coordinates suffice, RV3 clusters worse than RV2 (0.620 against 0.996 at $n$ = 100) because the unnecessary skewness coordinate injects estimation noise; and the added coordinates buy no insurance in the cells where skewness is also matched.

**TABLE 2** Clustering ARI at $n$ = 100 and $n$ = 2,000 (first and second entry in each cell) for the four worlds, with the true number of clusters supplied.

| Measure | Set 1 Gaussian | Set 2 Log-normal | Set 3 Mixture | Set 4 Extremes |
|---|---|---|---|---|
| $W_1$ | 0.995 / 1.000 | 0.642 / 1.000 | 0.526 / 1.000 | 0.355 / 0.999 |
| KS | 0.983 / 1.000 | 0.850 / 1.000 | 0.503 / 0.995 | 0.003 / 0.985 |
| SMD | 0.452 / 0.460 | 0.334 / 0.461 | 0.000 / 0.001 | 0.018 / 0.014 |
| RV1 | 0.489 / 0.495 | 0.343 / 0.494 | 0.000 / 0.001 | 0.026 / 0.023 |
| RV2 | 0.996 / 1.000 | 0.614 / 1.000 | 0.018 / 0.014 | 0.271 / 0.380 |
| RV3 | 0.620 / 0.736 | 0.448 / 0.953 | 0.131 / 0.339 | 0.189 / 0.247 |

ARI = 0 corresponds to chance agreement; 1 to exact recovery of the three true groups.

Supplying the true number of clusters is itself an oracle no sponsor has. Re-running the task with the number of clusters estimated by the silhouette criterion changes little for $W_1$ (its ARI at $n$ = 100 falls by 0.005–0.023 across the four worlds) and most for the KS statistic (by 0.260 in Set 1 and 0.179 in Set 2, because its distance structure leads the silhouette to choose two clusters). Two honest qualifications attach. First, in the hard worlds no measure recovers the true number of clusters at $n$ = 100: the modal choice is two for every measure in Sets 3 and 4, and the highest rate of choosing the true number is 0.228 and 0.288 respectively—a limit of the task, not a difference between measures. Nor is hitting the true number itself evidence of quality: in Set 3, measures whose ARI is near zero hit it at least as often as $W_1$, because a noise distance matrix can land on the right count by accident. Second, $W_1$'s Set 3 advantage over the KS statistic under the oracle (+0.023 at $n$ = 100) disappears without it (−0.005).

## 3.2 | Study 1: Estimation Properties

Study 2 compared decisions; Study 1 characterizes the estimator those decisions rely on: bias, variability, and coverage probability of bootstrap confidence intervals for $\widehat{W}_1$. These properties underwrite the clinical calibration framework, as reliable estimation of $W_1$ directly determines the quality of $\Delta_{\max}$ estimates via equation (8). We report operating characteristics in the original units of the effect modifier, with each scenario specified at $\sigma$ = 10; bias and CI width are therefore on the $W_1$ scale, while coverage is unitless and directly comparable across scenarios.

### 3.2.1 | Design

We designed seven systematic scenarios for methodological validation, examining controlled distributional differences across location, scale, and combinations thereof. Table 3 summarizes these scenarios with their true $W_1$ values, in the original units of the effect modifier.

**TABLE 3** Simulation scenarios with clinical motivations.

| ID | Description | Distribution 1 | Distribution 2 | True $W_1$ |
|---|---|---|---|---|
| S1 | Null (identical) | $N(50, 10^2)$ | $N(50, 10^2)$ | 0.00 |
| S2 | Location 0.2$\sigma$ | $N(50, 10^2)$ | $N(52, 10^2)$ | 2.00 |
| S3 | Location 0.5$\sigma$ | $N(50, 10^2)$ | $N(55, 10^2)$ | 5.00 |
| S4 | Location 1.0$\sigma$ | $N(50, 10^2)$ | $N(60, 10^2)$ | 10.00 |
| S5 | Scale 1.5× | $N(50, 10^2)$ | $N(50, 15^2)$ | 3.99 |
| S6 | Skew (Log-normal) | $N(50, 10^2)$ | LogN$(\mu, \sigma)$ | 12.15 |
| S7 | Location + Scale | $N(50, 10^2)$ | $N(55, 15^2)$ | 5.85 |

The Log-normal in S6 is parameterized to match mean 50 with CV ≈ 53%, typical of clinical biomarkers such as ALT. S7 combines location and scale differences, representing the most realistic MRCT pattern. True $W_1$ values for S1–S4 are exact ($W_1 = |\mu_1 - \mu_2|$ for equal-variance normals); S5 uses the closed form $W_1 = \sqrt{2/\pi}\,|\sigma_1 - \sigma_2|$; S6 and S7 are computed via Monte Carlo integration ($n_{\mathrm{MC}} = 10^6$).

For each scenario, we generated samples of size $n$ = 50, 100, and 200 per region, reflecting sample sizes typical in MRCT regional subgroups. We performed 10,000 replications per scenario–sample size combination to ensure stable estimates of operating characteristics, with Monte Carlo standard errors below 0.005 for all reported proportions. Bootstrap confidence intervals were computed using $B$ = 2,000 resamples. All simulations were conducted in R version 4.5.1.

Consistent with the estimation-centered framework, we evaluated the operating characteristics of the sample Wasserstein-1 estimator $\widehat{W}_1$ for the population value $W_1$, reported in the original units of the effect modifier:

1. Bias: $\text{Mean}(\widehat{W}_1) - W_1$
2. Root mean squared error (RMSE): $\sqrt{\text{Mean}[(\widehat{W}_1 - W_1)^2]}$
3. Coverage probability: Proportion of 95% bootstrap CIs containing the true value
4. CI width: Mean width of bootstrap CIs, reflecting estimation precision

### 3.2.2 | Results

Table 4 presents the bias of $\widehat{W}_1$ across scenarios and sample sizes, in the original units of the effect modifier (scenarios are specified with $\sigma = 10$). The estimator showed positive bias under the null hypothesis (S1), with bias decreasing from 2.46 at $n = 50$ to 1.26 at $n = 200$. This positive bias arises because the non-negativity of $W_1$ yields $\widehat{W}_1 > 0$ with probability 1 in finite samples, even when the true value equals zero ($F_1 = F_2$ almost everywhere). The estimator remains consistent ($\widehat{W}_1 \to 0$ as $n \to \infty$ under the null), but the convergence is slow near the parameter space boundary.

**TABLE 4** Bias of $\widehat{W}_1$ by scenario and sample size, in the original units of the effect modifier.

| Scenario | True $W_1$ | $n$ = 50 | $n$ = 100 | $n$ = 200 |
|---|---|---|---|---|
| S1 (Null) | 0.00 | 2.463 | 1.759 | 1.260 |
| S2 ($0.2\sigma$) | 2.00 | 0.990 | 0.472 | 0.173 |
| S3 ($0.5\sigma$) | 5.00 | 0.174 | 0.051 | 0.020 |
| S4 ($1.0\sigma$) | 10.00 | 0.034 | −0.009 | 0.004 |
| S5 (Scale) | 3.99 | 0.829 | 0.389 | 0.189 |
| S6 (Skew) | 12.15 | 0.382 | 0.210 | 0.116 |
| S7 (Loc+Scale) | 5.85 | 0.624 | 0.326 | 0.181 |

Bias is defined as $\text{Mean}(\widehat{W}_1) - W_1$, in the units of the effect modifier (scenarios specified with $\sigma = 10$). Results based on 10,000 replications with $B$ = 2,000 bootstrap resamples.

For the non-null scenarios away from the boundary (S3–S7), bias was small relative to each scenario's true $W_1$ and diminished with sample size; at $n = 200$ it was at most 0.189 (S5), a small fraction of the corresponding $W_1$ (Figure 3(A)). For S4 ($1.0\sigma$ location shift, $W_1 = 10$), bias was negligible (0.034 at $n = 50$, 0.004 at $n = 200$), confirming accurate estimation even for large distributional differences. For scenarios near the boundary (S1, S2), positive bias was more pronounced and diminished slowly with sample size—a well-known phenomenon for non-negative statistics where sampling variability inflates estimates when the true value is close to zero.

Table 5 presents coverage probabilities of the 95% bootstrap confidence intervals. For the non-null scenarios, coverage was near-nominal at $n \geq 100$: S3 (0.948–0.951), S4 (0.944–0.948), S6 (0.942–0.946), and S7 (0.936–0.945). S4 ($1.0\sigma$ location shift) showed stable coverage across all sample sizes, with no degradation at larger $n$, confirming satisfactory bootstrap performance even for large distributional differences. S5 (scale) showed moderate undercoverage at $n = 50$ (0.863) that improved to near-nominal at $n = 200$ (0.945). S2 (small location shift) showed undercoverage at $n = 50$ (0.703) due to its near-boundary true value, improving to 0.947 at $n = 200$.

Coverage for S1 (null) is shown as zero in Figure 3(B) because its true $W_1$ lies at the parameter space boundary. The non-negativity constraint produces persistent positive bias that prevents the finite-sample confidence interval from containing the true value of zero. This does not indicate estimator failure—the estimator is consistent—but rather that sample sizes well beyond 200 per region would be needed for reliable inference when the true distributional difference is negligible.

We also evaluated BCa (bias-corrected and accelerated) confidence intervals in preliminary experiments. BCa intervals showed consistently lower coverage than percentile intervals across all scenarios, particularly for near-boundary scenarios. The BCa

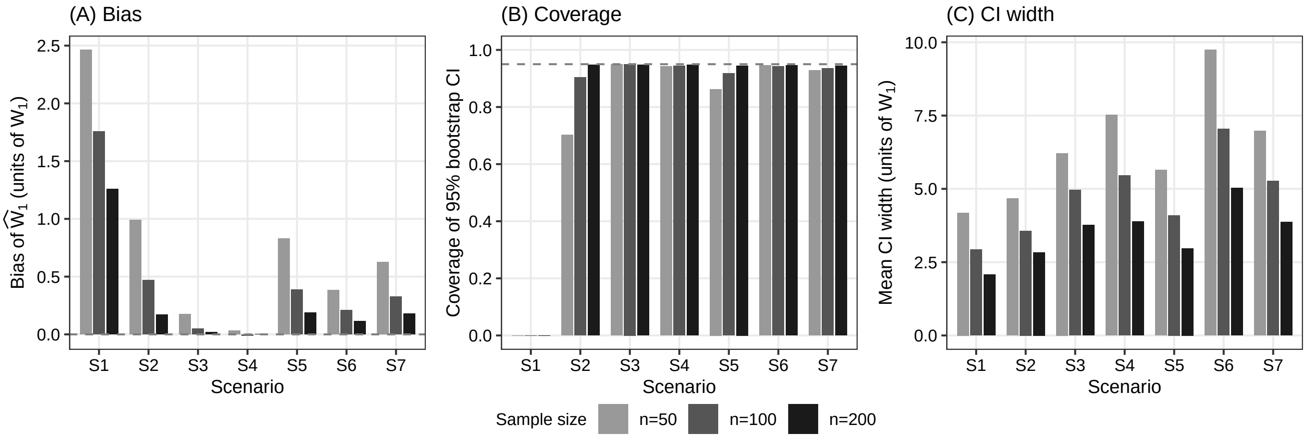


**FIGURE 3** Estimation properties of the sample Wasserstein-1 estimator $\widehat{W}_1$, in the original units of the effect modifier, across scenarios (S1–S7) and sample sizes ($n = 50, 100, 200$). Panel (A): bias, with dashed horizontal line at zero. Panel (B): coverage probability of 95% percentile bootstrap CIs, with dashed line at the nominal 0.95 level; S1 coverage is structurally zero because the true value lies at the parameter space boundary (see Section 3.2.2). Panel (C): mean CI width in units of $W_1$.

correction overcorrects for $\widehat{W}_1$ because the statistic is bounded below by zero, causing the acceleration parameter to distort the quantile adjustment. Based on these findings, we adopt percentile bootstrap as the primary inference method.

Monte Carlo standard errors for all reported coverage probabilities were below 0.005 at 10,000 replications, ensuring that observed differences across scenarios and sample sizes reflect genuine operating characteristic differences rather than simulation noise.

**TABLE 5** Coverage probability of 95% percentile bootstrap confidence intervals.

| Scenario | $n$ = 50 | $n$ = 100 | $n$ = 200 |
|---|---|---|---|
| S2 (0.2$\sigma$) | 0.703 | 0.904 | 0.947 |
| S3 (0.5$\sigma$) | 0.950 | 0.951 | 0.948 |
| S4 (1.0$\sigma$) | 0.942 | 0.944 | 0.948 |
| S5 (Scale) | 0.863 | 0.917 | 0.945 |
| S6 (Skew) | 0.946 | 0.942 | 0.946 |
| S7 (Loc+Scale) | 0.928 | 0.936 | 0.945 |

Coverage under S1 (Null) is shown as zero in Figure 3(B) but omitted from this table for brevity; the structural zero arises from the non-negativity constraint at the parameter space boundary.
For S4, coverage was near-nominal across all sample sizes (0.942–0.948), confirming satisfactory performance even for large distributional differences. The undercoverage for S2 at small sample sizes (0.703 at $n$ = 50) and S5 (0.863 at $n$ = 50) reflects boundary proximity effects that diminish with increasing $n$.

The mean CI width pattern across scenarios and sample sizes is shown in Figure 3(C). Table 6 presents the RMSE and mean CI width across scenarios, in the original units of the effect modifier. RMSE decreased consistently with increasing $n$, at the $O(n^{-1/2})$ rate expected from Wasserstein plug-in theory; for S3, for example, it fell from 1.83 at $n = 50$ to 0.98 at $n = 200$. Mean CI width narrowed proportionally, with CIs at $n = 100$ for the non-null scenarios spanning roughly 3.6–7.0 in units of $W_1$. In the clinical calibration framework, this CI width propagates directly to $\Delta_{\max}$ via $\Delta_{\max} = L_{\text{clinical}} \cdot W_1$ (Section 2.4). For example, with $L_{\text{clinical}} = 0.01$/yr (representative of age in the GUSTO-I application), a $\widehat{W}_1$ CI width of 4.96 years (S3 at $n = 100$) corresponds to a $\Delta_{\max}$ CI width of $0.01 \times 4.96 = 0.050$ (5.0%pt)—a level of precision sufficient for meaningful clinical calibration.

Under the null hypothesis (S1), the positive bias produces CIs that are shifted upward from zero. At $n = 50$, the mean estimate was 2.46 with mean CI width 4.19, and at $n = 200$, the mean estimate decreased to 1.26 with mean CI width 2.07. This bias is relevant for clinical calibration. When the true distributional difference is zero, the estimator will suggest a small positive $\Delta_{\max}$. Practitioners should be aware of this conservative tendency, particularly at smaller sample sizes. This boundary behaviour

**TABLE 6** RMSE and mean CI width of $\widehat{W}_1$ by scenario and sample size, in the original units of the effect modifier.

| Scenario | RMSE | | | Mean CI Width | | |
|---|---|---|---|---|---|---|
| | $n$ = 50 | $n$ = 100 | $n$ = 200 | $n$ = 50 | $n$ = 100 | $n$ = 200 |
| S1 (Null) | 2.63 | 1.87 | 1.34 | 4.19 | 2.94 | 2.07 |
| S2 ($0.2\sigma$) | 1.60 | 1.13 | 0.86 | 4.67 | 3.57 | 2.84 |
| S3 ($0.5\sigma$) | 1.83 | 1.37 | 0.98 | 6.22 | 4.96 | 3.77 |
| S4 ($1.0\sigma$) | 1.98 | 1.42 | 1.00 | 7.52 | 5.45 | 3.89 |
| S5 (Scale) | 1.64 | 1.11 | 0.77 | 5.65 | 4.10 | 2.97 |
| S6 (Skew) | 2.61 | 1.86 | 1.29 | 9.74 | 7.04 | 5.03 |
| S7 (Loc+Scale) | 2.00 | 1.44 | 1.01 | 6.97 | 5.27 | 3.87 |

CI width is defined as mean of (upper − lower) across 10,000 replications, in the units of the effect modifier (scenarios specified with $\sigma$ = 10).

is the operability phenomenon of Section 2.5 in its simplest instance: the null floor that diagnostic estimates is the sampling distribution characterized here.

In summary, the simulation study across seven scenarios demonstrates that the $\widehat{W}_1$ estimator provides reliable estimation with the following properties at sample sizes typical in MRCT regional subgroups:

1. Bias: Small relative to each scenario's $W_1$ at $n \geq 100$, with positive bias largest near the boundary (S1: 1.76 at $n$ = 100) and negligible for the larger-shift scenarios (S3: 0.051, S4: −0.009 at $n$ = 100).
2. Coverage: Near-nominal (0.92–0.95) at $n \geq 100$ for the non-null scenarios (S3–S7), including S6 (skew) and S7 (location + scale).
3. Precision: RMSE and CI width decrease at the $O(n^{-1/2})$ rate; at $n$ = 200, RMSE ranged from 0.77 (S5) to 1.34 (S1) in units of $W_1$.

Scenario S3 ($0.5\sigma$ location shift, true $W_1$ = 5) exemplifies the operating characteristics at clinically relevant effect sizes. Bias was negligible (+0.051 at $n$ = 100), coverage was nominal (0.951), and the CI width (4.96 years) translated to a $\Delta_{\max}$ CI width of $0.01 \times 4.96 = 0.050$ (5.0%pt) when calibrated with $L_{\text{clinical}}$ = 0.01/yr (age-scale parameters representative of the GUSTO-I application)—a level of precision sufficient for informed regulatory deliberation.

These estimation properties ensure that $\widehat{W}_1$, when combined with the clinical calibration framework of Section 2.4, provides $\Delta_{\max}$ estimates of sufficient quality for regulatory deliberation. We recommend $n \geq 100$ per region for reliable estimation and inference; this is the estimation criterion of the statements distinguished at the head of this section, and the sample sizes that decisions require are Study 2's (Section 3.1.3).

# 4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT USING GUSTO-I DATA

We illustrate the per-EM Wasserstein-1 framework through a hypothetical scenario grounded in publicly available data from GUSTO-I, a large international trial in acute myocardial infarction (AMI). A sponsor is developing a novel thrombolytic agent (Drug T) for AMI and is planning a Phase 3 MRCT across multiple regions. At the planning stage, the sponsor uses IPD from GUSTO-I ($N$ = 40,830; 16 regions)[20] to assess distributional similarity of candidate effect modifiers across regions and to identify suitable pooling partners for a designated anchor region. GUSTO-I was not designed as an MRCT and its regions are anonymized.

## 4.1 | Scenario Design and Effect Modifier Selection

For this illustrative exercise, the sponsor identifies two candidate effect modifiers for Drug T's treatment effect in AMI: age and systolic blood pressure (SBP). Drug T is a novel agent. Its Phase 2 programme has suggested age and SBP as candidate effect modifiers on the basis of exploratory analyses and biological plausibility, but the precision of these analyses is insufficient for numerical estimation of $L_{\text{clinical}}$. At this planning stage the sponsor therefore has no quantitative prior on the clinical slope $L_{\text{clinical}}$ for either candidate effect modifier, and class-level indirect evidence is of limited value as a numerical substitute. For age, the Fibrinolytic Therapy Trialists' (FTT) collaborative meta-analysis of existing fibrinolytic agents[19] concluded benefit is present irrespective of age and does not report a per-year CATE slope on the absolute mortality scale, so class transferability would not yield a defensible numerical prior for Drug T. For SBP, no class-level quantitative estimate of clinical slope is available at all. We therefore treat $L_{\text{clinical}}$ as unknown a priori for both candidate effect modifiers, and apply the $L^*$ reverse-calculation pathway (equation 9) to both.

The sponsor selects Region 8 as the anchor region—representing, for instance, a regulatory market where local efficacy evidence is required but sample size is limited. Region 8 comprises $n$ = 2,916 patients in GUSTO-I, with baseline characteristics summarized in Table 7. The exercise of identifying Region 8's suitable pooling partners simulates the decision a sponsor would face when seeking regional efficacy assessment through pooling as described in ICH E17.[3]

**TABLE 7** Region 8 baseline characteristics (GUSTO-I).

| Candidate effect modifier | Mean | SD | Skewness | $\text{IQR}_{\text{pooled}}$ |
|---|---|---|---|---|
| Age (years) | 60.2 | 12.1 | −0.17 | 17–18 |
| SBP (mmHg) | 132.4 | 22.9 | 0.20 | 30–34 |

$n$ = 2,916. $\text{IQR}_{\text{pooled}}$ ranges reflect variation across the 15 partner-region pairs.

## 4.2 | Operability Check: Is the Decision Resolvable at These Sample Sizes?

Before ranking partners it is worth asking whether the threshold can discriminate at all at the sample sizes available, applying the criterion of Section 2.5. For each candidate partner we drew two independent resamples, with replacement, from R8's own empirical distribution at sizes $n_{\text{R8}}$ = 2,916 and $n_{\text{partner}}$, recomputed $\widehat{W}_1$, and repeated this 2,000 times, so that the null floor is obtained separately for each partner as required by its dependence on both sample sizes. We take $\alpha$ = 0.05, and accordingly call a partner unresolved when its observed $\widehat{W}_1$ lies at or below the 95th percentile of that null distribution—that is, when the observed distance is one the estimator would produce with appreciable probability from two samples of identical distributions. Table 8 summarises the result.

**TABLE 8** Null floor of $\widehat{W}_1$ across the 15 candidate partners, by effect modifier. Ranges are across partners; the floor depends on each partner's sample size. Unresolved partners are those whose observed $\widehat{W}_1$ falls at or below the 95th percentile of the null distribution.

| Effect modifier | Null mean | Null 95th pct. | $\tau_{\text{clin}}$ | Unresolved |
|---|---|---|---|---|
| Age (years) | 0.359–0.510 | 0.618–0.900 | 1.0 | 6 / 15 |
| Systolic blood pressure (mmHg) | 0.704–1.013 | 1.176–1.691 | 5.0 | 1 / 15 |

The two effect modifiers behave differently, and the difference matters for how the subsequent rankings should be read. On age, six of the fifteen partners—R1, R4, R5, R7, R9 and R15—are unresolved, and all six fall on the eligible side of $\tau_{\text{clin}}$. For these partners an eligible verdict on age therefore does not establish similarity; it is equally consistent with a difference the data at hand cannot detect. On systolic blood pressure, by contrast, fourteen of fifteen are resolved, the single exception being R2. The decision is thus carried in substance by systolic blood pressure, and age contributes discrimination for only nine of the fifteen partners.

Because the null is drawn from one region's distribution, the choice of which region is not neutral: drawing from the anchor asks what $\widehat{W}_1$ would look like if the partner were distributed as R8 is, which is the null appropriate to an anchor-borrowing estimand. Repeating the computation with each partner's own distribution as the source leaves every verdict unchanged, so the finding does not depend on that choice.

This is not a report of failure. That $\widehat{W}_1$ has a null floor is a property of any non-negative distance, not a defect of this one, and the check exists precisely to identify which effect modifiers carry a given decision at a given sample size. Its practical consequence here is narrow but real: conclusions resting on age alone for the six unresolved partners would be conclusions the data do not support, and the joint criterion applied in Section 4.4 is what prevents that. It is also worth noting that the regional samples exceed two thousand observations—far above any per-region minimum motivated by bias and coverage in Section 3.2.2—which is what makes the point that operability is a separate question from estimation quality rather than a restatement of it.

## 4.3 | Distributional Assessment: Per-EM $W_1$ Across 15 Partner Regions

We computed $\widehat{W}_1$ between Region 8 and each of the remaining 15 regions for both candidate effect modifiers, expressed in the original measurement units (years for age, mmHg for SBP), with percentile bootstrap 95% confidence intervals ($B$ = 2,000). Table 9 presents the results. Figure 4 displays these results as a forest plot with bootstrap CIs, with partners ordered by ascending $\widehat{W}_{1,\text{age}}$ to aid visual comparison.

**TABLE 9** Per-EM Wasserstein-1 distance $\widehat{W}_1$ in original variable units (years for age, mmHg for SBP) with 95% percentile bootstrap CIs for Region 8 versus each partner region (GUSTO-I), listed by region number.

| Partner | $n$ | $\widehat{W}_{1,\text{age}}$ (yr) [95% CI] | $\widehat{W}_{1,\text{SBP}}$ (mmHg) [95% CI] |
|---|---|---|---|
| R1 | 2188 | 0.65 [0.51, 1.03] | 3.61 [2.68, 4.93] |
| R2 | 2952 | 2.15 [1.57, 2.76] | 0.95 [0.76, 1.82] |
| R3 | 2030 | 2.61 [1.96, 3.28] | 3.35 [2.32, 4.65] |
| R4 | 2876 | 0.56 [0.37, 1.16] | 2.60 [1.82, 3.75] |
| R5 | 1909 | 0.39 [0.35, 0.92] | 3.37 [2.32, 4.59] |
| R6 | 1585 | 0.91 [0.63, 1.39] | 3.37 [2.06, 4.77] |
| R7 | 3150 | 0.40 [0.30, 0.95] | 5.07 [4.03, 6.22] |
| R9 | 3123 | 0.59 [0.40, 1.12] | 6.80 [5.71, 8.01] |
| R10 | 1717 | 1.38 [0.79, 2.13] | 4.00 [2.88, 5.39] |
| R11 | 2491 | 1.27 [0.76, 1.91] | 6.44 [5.25, 7.71] |
| R12 | 4352 | 0.86 [0.47, 1.43] | 6.40 [5.39, 7.46] |
| R13 | 2297 | 1.15 [0.76, 1.74] | 2.40 [1.42, 3.72] |
| R14 | 3437 | 0.70 [0.42, 1.29] | 4.36 [3.41, 5.44] |
| R15 | 2576 | 0.61 [0.40, 1.25] | 4.72 [3.60, 5.94] |
| R16 | 1231 | 1.74 [1.16, 2.47] | 4.28 [3.06, 5.71] |

Point estimates with 95% percentile bootstrap CIs ($B$ = 2,000, seed = 2026). Regions are listed by region number, and ordering carries no implicit ranking. Bootstrap CIs quantify estimation uncertainty. Narrow CIs lend greater confidence to individual $\widehat{W}_1$ values, while overlapping CIs indicate partners whose relative position is not sharply resolved by the data.

The two candidate effect modifiers produce different patterns of distributional similarity. For age, partner-region $\widehat{W}_{1,\text{age}}$ point estimates span from 0.39 years (R5) at the low end to 2.61 years (R3) at the high end, with most partners below 1.5 years. For SBP, point estimates span from 0.95 mmHg (R2) to 6.80 mmHg (R9). Relative to their respective natural scales (age IQR $\approx$ 17–18 years and SBP IQR $\approx$ 30–34 mmHg in Region 8), the SBP differences are slightly larger in proportion than the age differences, and the ranking of partners by each candidate effect modifier differs accordingly.

Several partners fall in the middle of each ranking where uncertainty intervals overlap substantially. R16's $\widehat{W}_{1,\text{age}}$ (1.74 years) is the 13th-smallest value with 95% CI $[1.16, 2.47]$ years, placing it in a region where its relative position is not sharply determined by the data—the CI spans the range covered by several partners with smaller point estimates. R6's $\widehat{W}_{1,\text{age}}$ (0.91 years) is the 9th-smallest value, while its $\widehat{W}_{1,\text{SBP}}$ of 3.37 mmHg sits in the lower-middle of the SBP distribution with 95% CI $[2.06, 4.77]$ mmHg, similarly spanning a broad region of the SBP ranking. These mid-rank cases illustrate the value of reporting bootstrap CIs alongside point estimates. The ranking itself is accompanied by an uncertainty interval that informs how confidently one partner can be placed above or below another.

An instructive contrast emerges between R2 and R9. R2 has the second-largest $\widehat{W}_{1,\text{age}}$ (2.15 years) but the smallest $\widehat{W}_{1,\text{SBP}}$ (0.95 mmHg), while R9 has the 4th-smallest $\widehat{W}_{1,\text{age}}$ (0.59 years) but the largest $\widehat{W}_{1,\text{SBP}}$ (6.80 mmHg). A ranking based on either candidate effect modifier alone would reach opposite conclusions about these two regions. This asymmetry underscores the necessity of evaluating all candidate effect modifiers jointly.

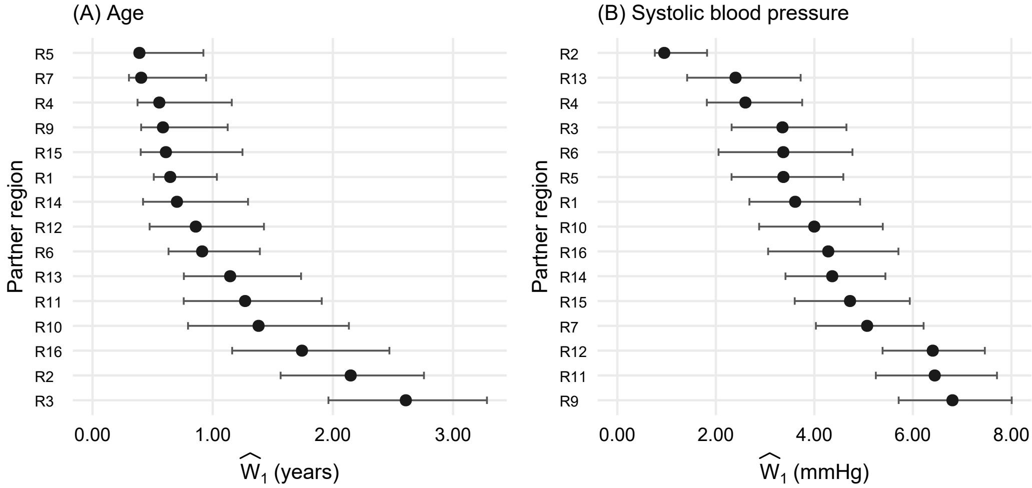


**FIGURE 4** Forest plots of $\widehat{W}_1$ point estimates in original units (years for age, panel A; mmHg for SBP, panel B) with 95% percentile bootstrap CIs. Partners are ordered by ascending $\widehat{W}_1$ within each panel; ordering and horizontal scale differ between panels.

## 4.4 | Clinical Interpretation and Pooling Candidates

We require an upper bound on the clinical slope $L_{\text{clinical}}$ for each candidate effect modifier; recall that $L_{\text{clinical}}$ governs how strongly treatment effect varies with the modifier. For thrombolysis in acute myocardial infarction, class-level meta-analytic evidence (GUSTO-I,[20] FTT collaborative meta-analysis,[19] and the GUSTO-I prognostic model[21]) supports a conservative upper bound of approximately 10%pt per decade for age-related and 2%pt per 10 mmHg for SBP-related treatment effect modification on 30-day mortality, exceeding the empirical prognostic gradients in Lee et al.[21] We adopt these as illustrative bounds: $L_{\text{age,UB}} = 1 \times 10^{-2}$ /yr and $L_{\text{SBP,UB}} = 2 \times 10^{-3}$ /mmHg.

For each partner region, we compare the region-specific $L^*$ value (Table 10) with the corresponding plausible upper bound $L_{\text{UB}}$. A region is judged eligible for pooling on a given effect modifier if $L^* > L_{\text{UB}}$, meaning the clinical slope required for the observed distributional difference to produce $\Delta_{\text{clin}} = 1$%pt exceeds the plausible bound. On age, the eligible partners are R1, R4, R5, R6, R7, R9, R12, R14, R15 (nine regions with $L^*_{\text{age}} > 0.01$). On SBP, the eligible partners are R1, R2, R3, R4, R5, R6, R10, R13, R14, R15, R16 (eleven regions with $L^*_{\text{SBP}} > 0.002$); R7 is excluded: its unrounded $L^*_{\text{SBP}} = 0.001972$ falls below $L_{\text{SBP,UB}} = 0.002$, although the rounded display in Table 10 shows 0.0020.

A region is jointly eligible if it satisfies the criterion on both candidate effect modifiers, since pooling requires distributional compatibility across all evaluated modifiers. Six regions meet this AND condition: R1, R4, R5, R6, R14, R15. The remaining nine partners fail on at least one effect modifier—for example, R2 fails on age ($L^*_{\text{age}} = 0.0047$), R9 fails on SBP ($L^*_{\text{SBP}} = 0.0015$)—and are not recommended for pooling under these clinical assumptions.

The robustness of this six-partner conclusion can be examined against its two inputs: the estimated distances and the clinical bounds. On the distance side, the margin separating each eligible partner from its threshold $\tau = \Delta_{\text{clin}}/L_{\text{UB}}$ (1.0 year for age, 5.0 mmHg for SBP) is unevenly distributed. Expressed as the relative slack $\tau/\widehat{W}_1 - 1$, four of the six jointly eligible regions clear both thresholds with slack of +14.6% or more (R5 reaches +156.5% on age), whereas the remaining two are each near one boundary: R6 sits +9.6% below the age threshold and R15 sits +5.8% below the SBP threshold. The fragility of the joint conclusion is therefore carried by R6 on age and R15 on SBP alone. The exclusion side is similarly narrow at its edge: R7 passes the age criterion comfortably ($\widehat{W}_{1,\text{age}} = 0.405$ years) and misses on SBP by 1.4% ($\widehat{W}_{1,\text{SBP}} = 5.072$ mmHg against the threshold of 5.000). The six–nine split is thus not a wide gap at its edges, and modest changes in either input can move individual regions across it.

On the clinical side, the bounds $L_{\text{age,UB}}$ and $L_{\text{SBP,UB}}$ are independent inputs from different evidence bases, so we vary each alone with the other held fixed; scaling both by a common factor would presume that two independently sourced clinical inputs err in the same direction by the same amount. Since $\tau = \Delta_{\text{clin}}/L_{\text{UB}}$, a larger bound is a stricter screen, and the slack values above are exactly the critical scaling factors: the age bound can rise by 9.6% before the first member (R6) drops, the SBP bound by

**TABLE** 10 Clinical interpretation ($L^*$ at $\Delta_{\text{clin}} = 1$%pt) and joint eligibility for all 15 partner regions, listed by region number. $L^*$ is the minimum clinical slope (per year for age, per mmHg for SBP) required for the observed distributional difference to produce a treatment effect difference of 1%pt on the absolute 30-day mortality scale. Eligibility is judged $L^* > L_{\text{UB}}$ with $L_{\text{age,UB}} = 1 \times 10^{-2}$ /yr and $L_{\text{SBP,UB}} = 2 \times 10^{-3}$ /mmHg.

| Partner | $\widehat{W}_{1,\text{age}}$ (yr) | $\widehat{W}_{1,\text{SBP}}$ (mmHg) | $L^*$ age (/yr) | $L^*$ SBP (/mmHg) | Age elig. | SBP elig. | Joint |
|---|---|---|---|---|---|---|---|
| R1 | 0.65 | 3.61 | 0.0154 | 0.0028 | ✓ | ✓ | ✓ |
| R2 | 2.15 | 0.95 | 0.0047 | 0.0105 | — | ✓ | — |
| R3 | 2.61 | 3.35 | 0.0038 | 0.0030 | — | ✓ | — |
| R4 | 0.56 | 2.60 | 0.0179 | 0.0038 | ✓ | ✓ | ✓ |
| R5 | 0.39 | 3.37 | 0.0256 | 0.0030 | ✓ | ✓ | ✓ |
| R6 | 0.91 | 3.37 | 0.0110 | 0.0030 | ✓ | ✓ | ✓ |
| R7 | 0.40 | 5.07 | 0.0247 | 0.0020 | ✓ | — | — |
| R9 | 0.59 | 6.80 | 0.0171 | 0.0015 | ✓ | — | — |
| R10 | 1.38 | 4.00 | 0.0072 | 0.0025 | — | ✓ | — |
| R11 | 1.27 | 6.44 | 0.0079 | 0.0016 | — | — | — |
| R12 | 0.86 | 6.40 | 0.0116 | 0.0016 | ✓ | — | — |
| R13 | 1.15 | 2.40 | 0.0087 | 0.0042 | — | ✓ | — |
| R14 | 0.70 | 4.36 | 0.0143 | 0.0023 | ✓ | ✓ | ✓ |
| R15 | 0.61 | 4.72 | 0.0164 | 0.0021 | ✓ | ✓ | ✓ |
| R16 | 1.74 | 4.28 | 0.0057 | 0.0023 | — | ✓ | — |

Regions are listed by region number, and ordering carries no implicit ranking. ✓ indicates eligibility ($L^* > L_{\text{UB}}$); — indicates ineligibility.

5.8% before R15 drops, and—because of R7's near-miss—the SBP bound needs to fall by only 1.4% for a seventh region to enter. Away from these edges the count degrades steadily. Varying $L_{\text{age,UB}}$ alone yields seven jointly eligible partners at 0.8 times the stated value, five at 1.1, four at 1.5, and one at 2; varying $L_{\text{SBP,UB}}$ alone yields seven at 0.8, five at 1.1, four at 1.25, one at 1.5 (R4 alone), and none at 2. The two bounds are not symmetric: at 1.5 times the stated value the age bound still retains four partners while the SBP bound retains one, so the joint conclusion rests principally on the SBP input.

This locates the sensitivity analysis and the operability check of Section 4.2 as two independent arguments for one conclusion. The operability check showed that SBP carries the decision because age is largely unresolved against its null floor—an estimation-resolution argument. The bound sensitivity shows that the conclusion is most fragile to the SBP clinical input—a clinical-input argument. Two different arguments identify the same effect modifier as the one on which the pooling decision actually rests.

Among the jointly eligible partners, R4 emerges as the leading single-pool candidate. R4 ranks 3rd-lowest on age ($\widehat{W}_{1,\text{age}} = 0.56$ years) and 3rd-lowest on SBP ($\widehat{W}_{1,\text{SBP}} = 2.60$ mmHg), balanced across both modifiers. R5 has the lowest age value ($\widehat{W}_{1,\text{age}} = 0.39$ years) but a mid-rank SBP value ($\widehat{W}_{1,\text{SBP}} = 3.37$ mmHg), an asymmetric profile. The framework supplies the quantitative inputs ($\widehat{W}_1$ with bootstrap CIs in Table 9, partner-specific $L^*$ in Table 10, and joint eligibility under the assumed bounds) to support sponsor judgment in consultation with clinical and regulatory advisors. Note that the GUSTO-I data were collected during 1990–1993; the application here serves as methodological illustration using publicly available IPD, not as an endorsement of these specific distributions for contemporary thrombolytic-trial planning.

## 4.5 | Comparative Application: Alternative Measures and Procedures

The preceding subsections applied the proposed framework alone. To place its selection in context, we repeated the complete partner-selection exercise with the alternative measures a sponsor could have used at the planning stage: the KS statistic and SMD of Section 2.1, and the three representative-value distances RV1–RV3 of Section 3.1.1 (RV1 the distance of Komiyama et al. as written; RV2 and RV3 our extensions).[6] KL divergence is omitted for the reasons given in Section 2.1. Because no alternative measure has a pathway from a clinical margin to a decision threshold, none can be given its own $\tau$; instead—mirroring Study 2's device—each measure admits its $k$ closest partners, where $k$ is the number $W_1$ admits at $\tau_{\text{clin}}$ ($k = 9$ on age, $k = 11$ on SBP). Matching the count removes any advantage or penalty from admitting more or fewer partners.

Table 11 summarizes the agreement of each measure with $W_1$. On both effect modifiers, the KS statistic, SMD, and RV1 admit exactly the partners $W_1$ admits, and the joint selection under the AND rule is identical for all four measures: R1, R4, R5, R6, R14, R15, precisely the six regions of the preceding subsection. Only RV2 and RV3 depart, and their joint selections differ from the consensus and from each other (RV2: R1, R4, R5, R7, R12, R15; RV3: seven regions, R1, R4, R7, R9, R10, R12, R15).

**TABLE 11** Agreement of each alternative measure with $W_1$ in the GUSTO-I application: overlap between the measure's $k$ closest partners and $W_1$'s ($k$ being the number of partners $W_1$ admits at $\tau_{\text{clin}}$), and Spearman rank correlation $\rho_S$ with $W_1$'s ordering of the 15 partners.

| EM | | KS | SMD | RV1 | RV2 | RV3 |
|---|---|---|---|---|---|---|
| age ($k$ = 9) | overlap | 9/9 | 9/9 | 9/9 | 8/9 | 8/9 |
| | $\rho_S$ | 0.964 | 0.896 | 0.896 | 0.911 | 0.904 |
| SBP ($k$ = 11) | overlap | 11/11 | 11/11 | 11/11 | 9/11 | 8/11 |
| | $\rho_S$ | 0.957 | 0.954 | 0.954 | 0.643 | 0.536 |

Overlap counts the shared members of the two $k$-partner sets; $\rho_S$ is computed on all 15 partners. RV1–RV3 are representative-value distances on the coordinates (mean), (mean, SD), and (mean, SD, skewness), respectively.

The mechanism of the agreement is that GUSTO-I's regional differences are predominantly differences in location. Where distributions differ mainly in their means, every measure that responds to the mean orders partners in nearly the same way. The configurations in which SMD is blind at any sample size—the moment-matched mixtures and mean-fixed displaced extremes of Study 2 (Section 3.1.2)—are constructible and clinically plausible, but they do not manifest in this dataset. What the comparison licenses is accordingly a controlled statement: in GUSTO-I, $W_1$ reaches the same selection as the existing measures, and its contribution here is not a different answer but the same answer plus its clinical calibration—the translation pathway from $\Delta_{\text{clin}}$ to a threshold, and the bound of equation (8), which no alternative carries. Since one cannot know before computing the distances whether one's own data are location-dominated, the agreement observed here cannot be presumed in advance.

Two further observations sharpen the comparison. First, RV1 and SMD produce identical partner orderings on both effect modifiers (Spearman correlation 1.000; Pearson 0.9999 on age and 0.9995 on SBP): on a single continuous effect modifier the representative value is the mean, so RV1 is a mean difference and SMD its standardized version. This is a scope observation rather than a criticism—the setting of the cited proposal describes regions by proportions of binary characteristics, where a single summary describes each distribution completely.[6] Second, the departures of RV2 and RV3 come with no guarantee of improvement: they are the only measures that leave the consensus, and their rank agreement with $W_1$ on SBP is markedly lower (0.643 and 0.536, against 0.954 for SMD and RV1). The instructive case is R6 on age, whose $\widehat{W}_{1,\text{age}} = 0.912$ years makes it the ninth-closest partner while its RV2 distance of 2.056 drops it from the admitted nine (displaced by R11), because the standard-deviation coordinate dominates the distance. Real data offer no ground truth against which to adjudicate this disagreement, but Study 2 does: adding moments is not free there (Section 3.1.4), and only the simulation and the application together license the reading that the departures of RV2 and RV3 are more likely noise than information. What the application alone shows is that augmenting the coordinate vector changes the selection, and moves it away from every other measure considered here.

The comparison so far varies the measure while holding the decision procedure fixed. Varying the procedure instead exposes the choice analyzed in Section 5: screening each candidate against the designated anchor, as in the preceding subsections, versus clustering the full pairwise distance matrix and reading off the anchor's cluster. On the same $W_1$ distances, the pairwise screen admits nine partners on age and eleven on SBP; cutting a complete-linkage dendrogram at height $\tau_{\text{clin}}$ leaves the anchor with two partners on each modifier (R5 and R6 on age; R2 and R16 on SBP); and clustering with a silhouette-selected number of clusters yields an anchor pool of ten partners on age and the same two-partner pool on SBP. The cut pool is a subset of the pairwise pool on both modifiers, as it must be, since a within-pool diameter of at most $\tau$ implies in particular that every anchor distance is at most $\tau$.

What separates the procedures is the diameter condition, and stating its scope precisely matters. For the stated purpose of this application—selecting pooling partners for the designated anchor—every anchor-to-partner distance in the admitted pools is within $\tau$ on the corresponding modifier, the anchor's heterogeneity bound (equation 6) holds, and nothing the method promises is breached. A diameter condition applies under a different reading, in which the pool is a shared pooled region whose every member, not only the anchor, carries the bound. The relevant quantity is then the largest pairwise $W_1$ within the pool, and its value depends on which pool is meant. For the six jointly eligible regions together with R8, the age diameter is 1.0595 years, exceeding $\tau_{\text{age}} = 1.0$ year by 5.95% on one of the fifteen partner pairs (R1–R6); the same pool's SBP diameter of 4.7243 mmHg is within its threshold. The single-modifier pools admitted by the pairwise screen stretch further: the nine-partner age pool has diameter 1.1093 years (10.9% above $\tau_{\text{age}}$), and the eleven-partner SBP pool has diameter 8.9362 mmHg (78.7% above $\tau_{\text{SBP}}$). The complete-linkage cut pools respect their thresholds by construction (diameters 0.9121 years and 4.2823 mmHg).

Two readings of these numbers are worth separating. The first is the cost of the extra condition: enforcing the shared-region reading reduces the admissible partners from nine to two on age and from eleven to two on SBP—a loss of 78% and 82% of the pool. That price is the practical reason the anchor estimand is the scope for which pairwise screening is built, and conversely why a pool intended as a shared pooled region calls for the diameter-controlled procedure. The second is a convergence: the breaching pair in the joint pool is R1–R6, and R6 is the same region the slack analysis of the preceding subsection flags as the age-borderline member, so two analyses single out one region. On SBP, by contrast, the joint pool's diameter is an anchor-to-partner distance rather than a partner pair, so on that modifier the anchor is the extreme point of its own pool and the shared-region reading imposes no additional cost.

Finally, alongside the distances we recorded for every pair the ratio $\rho = W_1/(D_{\mathrm{KS}} \cdot \sigma_{\mathrm{EM}})$, which measures the effective width, in SD units, of the range over which the two CDFs separate: $W_1$ integrates the CDF discrepancy whose supremum is $D_{\mathrm{KS}}$, so the two orderings can diverge only when discrepancies are spread over the support to differing degrees across pairs. In GUSTO-I the KS orderings track $W_1$ closely (Table 11), consistent with location-dominated differences. The role of $\rho$ must be bounded explicitly. It explains, after the fact, why the measures agreed in this dataset; it is not a pre-screening rule for deciding whether $W_1$ is needed, because $W_1$ appears in its numerator—a sponsor able to compute $\rho$ has already computed $W_1$. The quantity can say whether the KS statistic would have sufficed, never whether $W_1$ was needed.

## 5 | DISCUSSION

This paper addresses three methodological gaps identified in Section 2.1, demonstrating that the $W_1$ distance: (i) resolves distributional differences to which summary-based measures are structurally blind—in Study 2, every summary-based competitor (SMD and the representative-value distances) is at chance at every sample size examined in at least one clinically plausible world, an identification failure that no sample size repairs, whereas $W_1$ recovers the true partner structure in all four worlds (Section 3.1.2); (ii) translates distributional distance into a clinically interpretable scale via the heterogeneity bound $\Delta_{\max}$ (equation 8), a theoretical link to regional treatment effect heterogeneity that neither empirical-distribution-function statistics such as the Kolmogorov–Smirnov statistic nor representative-value distances possess;[6] and (iii) admits reliable percentile bootstrap inference for non-negligible distributional differences at moderate sample sizes ($n \geq 100$ per region), with a characterized boundary behavior near the null that the operability condition of Section 2.5 turns into a usable planning criterion.

These methodological findings carry two interpretive implications for pooling decisions, illustrated by the GUSTO-I application as a worked example of the framework's operation rather than a substantive epidemiological claim. First, when multiple candidate effect modifiers are under consideration, all candidates must be evaluated jointly. A partner ranked similar on one effect modifier may not be similar on another, as exemplified by the contrast between R2 and R9. Restricting evaluation to a subset of candidates risks selecting partners whose distributional differences on the omitted modifiers would later compromise consistency. Second, partner selection cannot rest on $W_1$ ranking alone. The required CATE sensitivity $L^*$ that would translate a given $W_1$ into a clinically meaningful $\Delta_{\max}$ varies markedly across partners and across candidate effect modifiers, so the same $W_1$ value carries different clinical weight depending on the partner-specific distributional differences. In the GUSTO-I example, six partners (R1, R4, R5, R6, R14, R15) emerged as jointly eligible because their $L^*$ values exceeded the plausible bounds reviewed in Section 4; R4 stood out as the leading single-pool candidate due to its balanced ranking on both modifiers. Pooling judgments therefore require both distributional ranking and clinical calibration, applied across the full set of candidate effect modifiers.

These implications also clarify how the framework relates to the most concrete existing pooling recipe, that of Komiyama et al., who combine multiple effect modifiers into a single Lasso-weighted distance and cluster regions on the resulting distance matrix.[6] The per-effect-modifier resolution adopted here matters precisely because partner rankings can reverse across modifiers: in the GUSTO-I example R2 and R9 each rank similar on one modifier but dissimilar on the other. This reversal is not anecdotal; across all 120 region pairs, the pairwise age and systolic blood pressure distances are nearly uncorrelated ($r = 0.13$), so the two effect modifiers carry almost independent distributional geometry, and collapsing them into one aggregate distance would systematically mask incompatibilities that the joint, per-modifier criterion of Section 4 retains. The two perspectives are nonetheless complementary, in two respects. First, the heterogeneity bound speaks directly to a question their account leaves open—whether a pooled-region treatment-effect estimate transfers to an individual member country. Within a pool whose members are mutually close on every candidate modifier, each pairwise regional difference in average treatment effect is bounded by $L_{\mathrm{clinical}} \cdot W_1$ (equation 6), so the pooled estimate applies to each member country with a clinically interpretable and bounded error, rather than resting on the point estimate of the pool alone. Second, our distance-based calibration answers which partners are similar enough, a question distinct from how many pooled regions to form; subgroup-multiplicity considerations suggest the latter should remain small, typically no more than four,[6,2] and the two constraints can be applied in sequence.

Which of the two decision procedures is appropriate—pairwise screening against a designated anchor region, as in Section 4, or clustering the full pairwise distance matrix, as in their account—is determined by the estimand rather than by statistical performance. When a single anchor region borrows strength from its partners—the setting of Section 4—screening candidates by their pairwise distance to the anchor is theoretically sufficient: $W_1$ is convex in mixtures of partner distributions, so the anchor-to-pool distance never exceeds the largest anchor-to-partner distance, and controlling each pairwise $W_1$ at a threshold $\tau$ controls the anchor's heterogeneity bound at $L_{\text{clinical}} \cdot \tau$. A supplementary simulation comparing the two procedures on a common outcome—the set of regions pooled with the anchor—found that pairwise screening then also performs better empirically, attaining higher sensitivity at matched control of the anchor's own violation rate in every simulated world and sample size examined. When instead pooled regions are formed at the planning stage and every member reports the shared pooled estimate—the setting contemplated by the pooled-region concept of ICH E17[3] and by Komiyama et al.[6]—pairwise screening under-protects: two partners each within $\tau$ of the anchor may lie up to $2\tau$ apart, so a member's own heterogeneity bound can reach twice the intended margin. In the same simulation this exposure was neither hypothetical nor a small-sample artefact: at a fixed clinical threshold, anchor-wise screening violated the pool-wide requirement in 61–68% of replications, and the rate did not fall with sample size across $n = 25$ to 400 per region, as expected of a structural rather than a sampling limitation. Cutting a complete-linkage dendrogram at height $\tau$ instead guarantees that every within-pool pairwise distance is at most $\tau$ in the estimated distances, so its residual breach reflects estimation error alone and declined from 16% at $n = 25$ to below 1% at $n = 400$. Clustering with a statistically selected number of clusters offered no such protection and frequently admitted dissimilar members, so the guarantee derives from the clinical cut, not from clustering as such. Procedure choice should therefore follow the estimand: pairwise screening for a single borrowing region, diameter-controlled clustering for shared pooled regions.

Section 2.1 identified three gaps in existing distributional measures: SMD captures only location; the KS statistic and related EDF tests lack a clinically interpretable scale and a theoretical link to treatment effect heterogeneity; and KL divergence is asymmetric, can diverge under non-overlapping support, and requires density estimation impractical at small regional samples. The $W_1$ distance addresses all three. The location-only gap is structural, and Study 2 exhibits its decision-level consequence: in the moment-matched worlds, SMD is at chance at every sample size examined while $W_1$ recovers the partner structure (Section 3.1.2). The clinical-scale gap is closed by the heterogeneity bound (equation 8), which translates distributional distance into clinically meaningful units. The instability gap is closed by construction: $W_1$ is symmetric, finite for any pair of empirical distributions, and computable directly from CDFs without density estimation, while remaining amenable to bootstrap inference. Addressing the gaps does not, however, mean returning a different answer in every dataset. In GUSTO-I, whose regional differences are predominantly differences in location, $W_1$ reached the same partner selection as the existing measures, and its contribution there was the clinical calibration rather than a different answer (Section 4.5). The gaps bind where distributional differences go beyond location—mixtures and rare displaced subgroups, configurations that are clinically ordinary—and which kind of dataset one holds is not knowable before the distances are computed. The reason to use $W_1$ is therefore not that it always gives a different answer, but that it is the only measure examined that both survives the worlds where the answer changes and carries a translation to the clinical scale.

Two features of the framework bear on its use in practice. First, the calibration adapts to where a trial sits on the spectrum of CATE sensitivity evidence rather than imposing a fixed $W_1$ cutoff: at the planning stage, where $L$ is typically unavailable for a novel agent, the $L^*$ reverse-calculation (equation 9) is the primary calibration tool; when $L$ is available from prior evidence, the $\Delta_{\max}$ pathway (equation 8) provides a complementary calibration on the clinical scale. Second, the distributional assessment and its calibration are separable. The identification advantage of the preceding paragraph is a property of the distance itself and holds whether or not $L$ is available; clinical calibration enhances but does not replace it.

For practice, we recommend computing $W_1$ with bootstrap CIs at moderate sample sizes (Section 3) for every candidate effect modifier across region pairs, translating each into $\Delta_{\max}$ (when $L$ is estimable) or $L^*$ at pre-specified $\Delta_{\text{clin}}$ values (when $L$ is unknown), and reporting these alongside clinically relevant benchmarks (overall treatment effect, non-inferiority margin) to support deliberation rather than binary decisions. When multiple candidate effect modifiers are evaluated, sponsors may adopt a conservative approach based on the maximum $\Delta_{\max}$ (or smallest $L^*$) across all effect modifiers and region pairs, or a totality-of-evidence approach in which the full collection—together with bootstrap uncertainty and the reliability of each $L$ estimate—informs holistic judgment. The choice depends on regulatory context.

For policy, the framework's compatibility with diverse data sources is a distinctive practical advantage. Because $W_1$ requires only baseline effect modifier distributions, regional data can be assembled from prior trials, registries, electronic health records, or other real-world evidence (RWE) sources—an increasingly relevant capability as regulatory agencies promote RWE use in drug development (ICH E6(R3)). The framework is particularly relevant for regulatory submissions requiring regional subpopulation consistency.[4,5] Matsushima et al.'s[4] PMDA workshop documented how regional imbalance in CRP-positive/MRI-negative status caused apparent inconsistency in the secukinumab MRCT, underscoring the practical consequences when effect modifier distributional differences are not assessed prospectively.

The $W_1$ distance is one input within a holistic regulatory evaluation; biological plausibility, clinical relevance, and benefit–risk balance extend beyond what any single distributional measure can quantify.[3,4] The framework has further limitations. It treats continuous effect modifiers marginally, leaving categorical and multivariate extensions for future work. Being non-negative, $W_1$ exhibits positive bias and zero bootstrap coverage when the true value lies at or near zero (S1; Section 3.2.2),[24] so we recommend $n \geq 100$ per region and cautious interpretation of estimates near the null. Clinical calibration depends on the CATE sensitivity $L$, which may be unavailable for novel agents (handled by the $L^*$ pathway) or transferred from prior evidence (a clinical assumption); the framework supplies quantitative inputs but prescribes no cutoffs. Finally, as with any covariate-based approach, unmeasured effect modifiers may contribute to heterogeneity beyond what the index captures.

Beyond the scope and methodological extensions noted above, two further directions warrant investigation. The upstream identification of which baseline characteristics constitute relevant effect modifiers is itself a non-trivial problem deserving dedicated treatment; regression-based relevance weighting such as the Lasso offers one front-end for this selection,[6] after which the present framework quantifies distributional distance on the retained modifiers. Bias-correction methods tailored to the boundary behavior of $W_1$ at small samples could extend the operational range of inference. The principal contribution of this paper is to supply a standardized, clinically calibrated tool where none was previously available. The $W_1$ distance, combined with bootstrap confidence intervals and clinical calibration via $\Delta_{\max}$ or $L^*$, transforms the previously qualitative judgment of "similar enough" into a reproducible, quantitative basis for sponsor judgment, filling the methodological gap left by ICH E17 and enabling evidence-based, clinically grounded pooling decisions.

## AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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## FINANCIAL DISCLOSURE

None reported.

## CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

## DATA AVAILABILITY STATEMENT

R code for computing the Wasserstein-1 distance and reproducing the simulation study is available at [repository URL]. GUSTO-I individual patient data are available from the original investigators subject to applicable data sharing policies; the dataset is described in the original publication.[20]

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## SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website. Supporting materials include: complete R code for Wasserstein-1 distance computation, simulation scripts, and additional figures for realistic clinical scenarios.

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## APPENDIX

## A PROOFS AND MATHEMATICAL DETAILS

### A.1 Non-negativity and Identity of Indiscernibles

Non-negativity is immediate: $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx \geq 0$, since the integrand is non-negative.

For the identity of indiscernibles, we show both directions. If $F_1 = F_2$, then $|F_1(x) - F_2(x)| = 0$ for all $x$, so $W_1(F_1, F_2) = 0$. Conversely, if $W_1(F_1, F_2) = \int |F_1(x) - F_2(x)|\, dx = 0$, then since $|F_1(x) - F_2(x)| \geq 0$ and is right-continuous, the integral can vanish only if $F_1(x) = F_2(x)$ almost everywhere, and by right-continuity of CDFs, $F_1 = F_2$ everywhere.

### A.2 Asymptotic Properties

Under standard regularity conditions, the empirical Wasserstein-1 distance $\widehat{W}_1 = W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$.[12]

In one dimension, the empirical $W_1$ distance converges at rate $O(n^{-1/2})$ without logarithmic correction.[12] For two-sample estimation with $F_1 \neq F_2$ and finite second moments,

$$\sqrt{\tfrac{n_1 n_2}{n_1+n_2}}\left(\widehat{W}_1 - W_1(F_1, F_2)\right) \xrightarrow{d} N(0,\, \sigma^2_{W_1}), \tag{A1}$$

where $n_1/(n_1 + n_2) \to \lambda \in (0, 1)$ and $\sigma^2_{W_1}$ depends on $F_1$ and $F_2$ through the limiting two-sample empirical process.[12] This result requires $W_1 > 0$ (i.e., $F_1 \neq F_2$). At the boundary $F_1 = F_2$, the parameter lies on the edge of the parameter space and standard asymptotics do not apply, consistent with the degenerate bootstrap coverage observed under the null (S1; see Section 5).

The bootstrap provides valid inference for the Wasserstein distance under mild conditions.[14] The percentile bootstrap confidence interval achieves asymptotically correct coverage for non-degenerate distributions.

## B R CODE FOR $W_1$ COMPUTATION

**Listing 1** R function for computing the Wasserstein-1 distance with bootstrap confidence intervals.

```
compute_W1 <- function(x1, x2) {
  pooled <- c(x1, x2)
  all_vals <- sort(unique(pooled))
  F1 <- ecdf(x1); F2 <- ecdf(x2)
  w1 <- sum(diff(all_vals) *
    abs(F1(all_vals[-length(all_vals)]) -
        F2(all_vals[-length(all_vals)])))
  w1
}

W1_bootstrap <- function(x1, x2,
    B = 2000, conf = 0.95) {
  obs <- compute_W1(x1, x2)
  boot_vals <- replicate(B, {
    b1 <- sample(x1, replace = TRUE)
    b2 <- sample(x2, replace = TRUE)
    compute_W1(b1, b2)
  })
  alpha <- 1 - conf
  ci <- quantile(boot_vals,
    c(alpha/2, 1 - alpha/2), na.rm = TRUE)
  list(estimate = obs,
       ci_lower = ci[1], ci_upper = ci[2])
}
```